WORD-COMPATIBLE GRAPHS AND GROVE POLYNOMIALS

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Abstract.

1. WC graphs

Definition 1.1. A *WC graph* is simply a finite subset $W \subseteq \mathbb{N} \times \mathbb{N}$ with no additional restrictions. We impose a total ordering on these pairs by declaring that $(a, b) \leq (c, d)$ if $a < c$ or $a = c$ and $b \geq d$. We define the *weight* of a WC graph W to be the sequence

$$\mathtt{wt}(W) = (i_1, \dots, i_n)$$

where i_j is the number of elements of W in row j. We define the *permutation* of a WC graph W to be the permutation

$$\mathtt{perm}(W) = s(w_1) \bullet \cdots \bullet s(w_n)$$

where

$$w_i = (a_i, b_i)$$

is the ith element in the imposed ordering, and

$$s(a_i, b_i) = s_{a_i + b_i - 1}$$

with $\bullet$ being the Hecke product. The *word* of W is

$$(s(w_1), \dots, s(w_n))$$

Define $\mathcal{WC}(w)$ to be the set of all WC graphs W such that $\mathtt{perm}(W) = w$.

The following is well-known.

Theorem 1.2. *We have*

$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{W \in \mathcal{WC}(w)} \beta^{|W| - \ell(w)} x^{\mathtt{wt}(W)}$$

2. Grove polynomials and WC graphs

Definition 2.1. A labeled indexed forest (F, P) is an indexed forest F with a function (labeling) $P : \mathrm{IN}(F) \to \mathbb{N}$ satisfying the following properties:

(1) $P(\mathrm{left}(u)) < P(u)$ for all $u \in \mathrm{IN}(F)$.
(2) $P(u) \leq P(\mathrm{right}(u))$ for all $u \in \mathrm{IN}(F)$.
(3) $P(u) \geq \rho_F(u)$ for all $u \in \mathrm{IN}(F)$.
(4) $P(u) \leq \lambda_F(u)$ for all $u \in \mathrm{IN}(F)$.

The set $\mathrm{Komp}_P(F)$ is the set of functions $\kappa : \mathrm{IN}(F) \to 2^{\mathbb{N}}$ such that

(1) $\max(\kappa(u)) \leq P(u)$ for all $u \in \mathrm{IN}(F)$.
(2) $\max(\kappa(u)) \leq \min(\kappa(\mathrm{left}(u)))$ for all $u \in \mathrm{IN}(F)$.
(3) $\max(\kappa(u)) < \min(\kappa(\mathrm{right}(u)))$ for all $u \in \mathrm{IN}(F)$.

We have that

$$\mathtt{wt}_i(\kappa) = \#\{u \in \mathrm{in}(F) \mid i \in \kappa(u)\}$$

Theorem 2.2. *Suppose a labeled indexed forest (F,P) is such that there is a $\kappa \in \mathrm{Komp}_P(F)$ with*

$$\mathtt{wt}(\kappa) = \mathfrak{c}(F).$$

Then

$$\widetilde{\mathfrak{P}}_F(x) = \sum_{\kappa \in \mathrm{Komp}_P(F)} \beta^{|\kappa|-|F|} x^{\mathtt{wt}(\kappa)}$$

Definition 2.3. Let $W \in \mathcal{WC}(w)$ be a WC graph for w. Define via the Nadeau-Tewari insertion algorithm

$$(P(W), Q(W)) = (P(\mathrm{word}(W)), Q(\mathrm{word}(W)))$$

and let

$$F(W) = \mathrm{shape}(P(W))$$

Define $\mathfrak{c}(F)$ to be the code of the indexed forest F.

Definition 2.4. Let (F,P) be a labeled indexed forest and let $\kappa \in \mathrm{Komp}_P(F)$. Consider the set

$$A = \bigcup_{v \in \mathrm{IN}(F)} \kappa(v) \times \{P(v)\}$$

Define

$$\mathcal{WC}(\kappa) = \{(a_i, b_i + 1 - a_i) \mid (a_i, b_i) \in A\}$$

Theorem 2.5. *For a permutation w, let $\mathcal{WC}_{\mathfrak{P}}(w)$ be the set of $W \in \mathcal{WC}(w)$ such that $\mathtt{wt}(W) = \mathfrak{c}(F(W))$. Then we have that $\mathcal{WC}(w)$ is the disjoint union of $\mathcal{WC}(\mathrm{Komp}_{P(W)}(F(W)))$ for $W \in \mathcal{WC}_{\mathfrak{P}}(w)$.*

Corollary 2.6. *The Grothendieck polynomial $\mathfrak{G}_w^{(\beta)}(x)$ satisfies*

$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{F \in \mathrm{IF}} \beta^{|F|-\ell(w)} a_w^F \widetilde{\mathfrak{P}}_F(x)$$

where

$$a_w^F = \#\{W \in \mathcal{WC}_{\mathfrak{P}}(w) \mid F(W) = F\}$$

Theorem 2.7. *Let a be a composition. Let w be the permutation such that $\mathfrak{c}(w) = a$. Then we have that*

$$\widetilde{\mathfrak{P}}_a(x) = \sum_{W \in \mathfrak{P}\mathcal{WC}(a) \cap \mathcal{WC}_{\mathrm{QY}}(w)} \mathcal{G}_{\mathtt{wt}(W)}(x)$$

3. Hecke insertion, K-Knuth equivalence, and Lascoux classes

The purpose of this section is to attach to every WC graph a *Lascoux class*, namely the K-Knuth equivalence class of the increasing tableau obtained by Hecke insertion of its word. The main input is a theorem, due to Reiner–Yong [5] (conjecture) and Shimozono–Yu [7] (proof), stating that the WC graphs of w lying in a single K-Knuth class have generating function equal to a single Lascoux polynomial. This refines the symmetric statement of Buch–Kresch–Shimozono–Tamvakis–Yong [1]. We phrase everything in the WC-graph language of the previous sections, so that it can be used to show that WC-graph transition, and hence the squash product, preserves Lascoux classes.

3.1. **The diagonal word of a WC graph.** Recall from the definition of a WC graph that the elements of W are listed as $w_1 < w_2 < \cdots < w_n$ in the total order on $\mathbb{N} \times \mathbb{N}$, with $w_i = (a_i, b_i)$, and that

$$s(a_i, b_i) = s_{a_i+b_i-1}.$$

Definition 3.1. The *diagonal word* of a WC graph W is the word of positive integers

$$\mathtt{dw}(W) = (a_1 + b_1 - 1,\ a_2 + b_2 - 1,\ \ldots,\ a_n + b_n - 1),$$

read in the total order on W. Thus $\mathrm{word}(W) = (s_{c_1}, \ldots, s_{c_n})$ where $\mathtt{dw}(W) = (c_1, \ldots, c_n)$, and

$$\mathtt{perm}(W) = s_{c_1} \bullet s_{c_2} \bullet \cdots \bullet s_{c_n}$$

is the Hecke product of the letters of $\mathtt{dw}(W)$.

3.2. **Hecke insertion and K-Knuth equivalence.** We use *Hecke row insertion* in the sense of Buch–Kresch–Shimozono–Tamvakis–Yong [1]. Recall that an *increasing tableau* is a filling of a straight (Young-diagram) shape by positive integers that is strictly increasing along rows and down columns. Hecke insertion of a word $c = (c_1, \dots, c_n)$ produces a pair

$$\big(\mathsf{P}(c), \mathsf{Q}(c)\big),$$

where $\mathsf{P}(c)$ is an increasing tableau (the *insertion tableau*) and $\mathsf{Q}(c)$ is a set-valued standard recording tableau of the same shape, obtained by inserting $c_1, \dots, c_n$ in order; each insertion either creates one new box or leaves the shape unchanged, the latter being the K-theoretic feature. For a WC graph W we write

$$\mathsf{P}(W) = \mathsf{P}(\mathtt{dw}(W)), \qquad \mathsf{Q}(W) = \mathsf{Q}(\mathtt{dw}(W)).$$

We record the three properties we use; all are due to [1] and [2].

Theorem 3.2 ([1])**.** *Hecke insertion $c \mapsto (\mathsf{P}(c), \mathsf{Q}(c))$ is a bijection from words of positive integers to pairs (P, Q) in which P is an increasing tableau and Q is a set-valued standard recording tableau of the same shape.*

Definition 3.3. The *K-Knuth equivalence* $\overset{K}{\equiv}$ is the equivalence relation on words of positive integers generated by the following moves, in which $\mathbf{u}$ and $\mathbf{v}$ are arbitrary words and a, b, c are positive integers:

$$(\mathbf{u}, a, a, \mathbf{v}) \overset{K}{\equiv} (\mathbf{u}, a, \mathbf{v}), \tag{3.2.1}$$

$$(\mathbf{u}, a, b, a, \mathbf{v}) \overset{K}{\equiv} (\mathbf{u}, b, a, b, \mathbf{v}), \tag{3.2.2}$$

$$(\mathbf{u}, a, b, c, \mathbf{v}) \overset{K}{\equiv} (\mathbf{u}, a, c, b, \mathbf{v}) \qquad \text{if } b < a < c, \tag{3.2.3}$$

$$(\mathbf{u}, a, b, c, \mathbf{v}) \overset{K}{\equiv} (\mathbf{u}, b, a, c, \mathbf{v}) \qquad \text{if } a < c < b. \tag{3.2.4}$$

The last two moves are the K-theoretic Knuth relations and the first two are the idempotency and braid relations of the 0-Hecke monoid. In particular K-Knuth equivalence refines Hecke equivalence: if $c \overset{K}{\equiv} c'$ then $s_{c_1} \bullet \cdots \bullet s_{c_n} = s_{c'_1} \bullet \cdots \bullet s_{c'_m}$.

Theorem 3.4 ([1, 2])**.** *For words c, c' of positive integers the following hold.*

1. *$\mathsf{P}(c) = \mathsf{P}(c')$ if and only if $c \overset{K}{\equiv} c'$; thus the insertion tableau is a complete invariant of the K-Knuth class.*
2. *The Hecke product $s_{c_1} \bullet \cdots \bullet s_{c_n}$ depends only on the K-Knuth class of c; it equals the Hecke product of the row reading word of $\mathsf{P}(c)$.*

Definition 3.5. The *Lascoux class* of a WC graph W is the K-Knuth equivalence class of $\mathtt{dw}(W)$, equivalently the insertion tableau $\mathsf{P}(W)$. We write

$$[W] = \big[\mathtt{dw}(W)\big]_K = \mathsf{P}(W).$$

By Theorem 3.4(2), every W in a fixed Lascoux class has the same permutation $\mathtt{perm}(W)$; we call it the permutation of the class.

3.3. **Lascoux polynomials.** Let ∂_i denote the ith divided difference operator,

$$\partial_i f = \frac{f - s_i f}{x_i - x_{i+1}},$$

and define the *K-theoretic isobaric (Demazure) operator*

$$\pi_i^{(\beta)} f = \partial_i\big(x_i(1 + \beta x_{i+1}) f\big).$$

Then $\pi_i^{(\beta)}(1) = 1$, $\pi_i^{(\beta)}$ sends polynomials to polynomials whose value is symmetric in x_i, x_{i+1}, and at $\beta = 0$ it recovers the Demazure operator $\pi_i f = \partial_i(x_i f)$ of the key polynomials.

Definition 3.6. For a weak composition $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ the *Lascoux polynomial* $\mathfrak{L}_\alpha$ is defined by:

1. if $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$ (i.e. α is a partition), then $\mathfrak{L}_\alpha = x^\alpha = \prod_i x_i^{\alpha_i}$;
2. if $\alpha_i < \alpha_{i+1}$ for some i, then $\mathfrak{L}_\alpha = \pi_i^{(\beta)} \mathfrak{L}_{s_i\alpha}$, where $s_i\alpha$ interchanges α_i and α_{i+1}.

This is well defined by [7]. The lowest-β component of $\mathfrak{L}_\alpha$ (its terms of total degree $|\alpha|$, i.e. the β^0 part) is the key (Demazure) polynomial κ_α.

3.4. **The Grothendieck-to-Lascoux expansion.** The next theorem is the symmetric statement of Buch–Kresch–Shimozono–Tamvakis–Yong, in the WC-graph normalization: grouping the WC graphs of w by Lascoux class collects their generating function into symmetric Grothendieck building blocks.

Theorem 3.7 ([1]). *Fix an increasing tableau P of straight shape λ whose row reading word has Hecke product w. Then*

$$\sum_{\substack{W\in\mathcal{WC}(w)\\ \mathsf{P}(W)=P}} \beta^{|W|-\ell(w)} x^{\mathtt{wt}(W)}$$

is symmetric in the variables it involves; summed over all such P of a fixed shape λ it equals the stable (Grassmannian) Grothendieck contribution $a_{w,\lambda}\, G_\lambda$, recovering the factor-sequence expansion $G_w = \sum_\lambda a_{w,\lambda} G_\lambda$.

The nonsymmetric refinement replaces each symmetric block by a single Lascoux polynomial. This is the "Grothendieck to Lascoux" expansion.

Theorem 3.8 (Reiner–Yong [5]; Shimozono–Yu [7]). *There is a map $P \mapsto \alpha(P)$ assigning to each increasing tableau P a weak composition $\alpha(P)$, with underlying partition $\mathrm{sh}(P)$, determined by the K-theoretic right key of P, such that for every increasing tableau P arising as an insertion tableau of some $W \in \mathcal{WC}(w)$,*

$$\sum_{\substack{W\in\mathcal{WC}(w)\\ \mathsf{P}(W)=P}} \beta^{|W|-\ell(w)} x^{\mathtt{wt}(W)} = \mathfrak{L}_{\alpha(P)}.$$

Consequently

$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{P\in\mathcal{I}(w)} \mathfrak{L}_{\alpha(P)} = \sum_{\alpha} c_{w,\alpha}\, \mathfrak{L}_\alpha, \qquad c_{w,\alpha} = \#\{P \in \mathcal{I}(w) \mid \alpha(P) = \alpha\} \in \mathbb{Z}_{\geq 0},$$

where $\mathcal{I}(w) = \{\mathsf{P}(W) \mid W \in \mathcal{WC}(w)\}$ is the set of Lascoux classes of w.

The content we use below is the following: *the Lascoux polynomial attached to a WC graph depends only on its Lascoux class.* We isolate this as a corollary.

Corollary 3.9. *The assignment*

$$W \longmapsto \mathfrak{L}_{\alpha(\mathsf{P}(W))}$$

factors through the Lascoux class $[W] = \mathsf{P}(W)$. In particular, if Φ is any map on WC graphs with $\mathsf{P}(\Phi(W)) \overset{K}{\equiv} \mathsf{P}(W)$ for all W, equivalently $\mathtt{dw}(\Phi(W)) \overset{K}{\equiv} \mathtt{dw}(W)$, then

$$\mathfrak{L}_{\alpha(\mathsf{P}(\Phi(W)))} = \mathfrak{L}_{\alpha(\mathsf{P}(W))}$$

for all W; that is, Φ preserves Lascoux classes and hence the associated Lascoux polynomial.

Proof. Immediate from Theorem 3.4(1), which identifies $\mathsf{P}(W)$ with the K-Knuth class of $\mathtt{dw}(W)$, and Theorem 3.8, in which α is a function of $\mathsf{P}(W)$ alone. □

Remark. Corollary 3.9 is the K-theoretic, straight-shape analogue of the grove/forest decomposition of Section 3's predecessor: there the WC graphs of w are grouped by the indexed forest $F(W)$ of the Nadeau–Tewari insertion and each fiber contributes $\widetilde{\mathfrak{P}}_{F(W)}$, whereas here they are grouped by the K-Knuth class $\mathsf{P}(W)$ of Hecke insertion and each fiber contributes $\mathfrak{L}_{\alpha(\mathsf{P}(W))}$. Identifying the two groupings, e.g. establishing an equality $\widetilde{\mathfrak{P}}_F = \mathfrak{L}_{\alpha(\mathsf{P})}$ under the correspondence $F \leftrightarrow \mathsf{P}$, would identify grove polynomials with Lascoux polynomials.

4. Bounded WC graphs and the K-theoretic transition

We now equip WC graphs with a row grading and a transition map, mirroring the treatment of bounded RC graphs in the reduced theory of [6]. Everything in this section specializes to those reduced constructions on the subset of RC graphs.

Definition 4.1. A *bounded WC graph* is a WC graph W together with a specified number of rows $n \geq \max\{a : (a,b) \in W \text{ for some } b\}$ subject to $\mathtt{m}(\mathtt{perm}(W)) \leq n$. We write (W,n) for the pair, abuse notation to treat the number of rows as a property of W, and call $\mathtt{ht}(W) = n$ the *height* of W. For $\mathtt{ht}(W) \geq 1$ we define the *trim* $\mathtt{trim}(W)$ to be the bounded WC graph with height $\mathtt{ht}(W) - 1$ and underlying set

$$\mathtt{trim}(W) = \{(a-1, b-1) : (a,b) \in W,\ a \geq 2\},$$

and we set $\uparrow^m W = \{(a+m, b) \mid (a,b) \in W\}$; these are the bounded RC-graph operations of [6] applied to WC graphs verbatim. Let $\mathcal{BWC}$ be the free abelian group on all bounded WC graphs, graded by height,

$$\mathcal{BWC} = \bigoplus_{n=0}^{\infty} \mathcal{BWC}_n.$$

A WC graph with $\ell(\mathtt{perm}(W)) = |W|$ is exactly an RC graph, and on such graphs $\mathtt{trim}$ and $\uparrow^m$ agree with the RC operations. Thus $\mathcal{BRC} \subseteq \mathcal{BWC}$ is a graded subgroup and the projection $\rho : \mathcal{BWC} \to \mathcal{BRC}$ that kills every WC graph with $|W| > \ell(\mathtt{perm}(W))$ is a graded retraction.

4.1. **The transition map via marked bumpless pipe dreams.** In the reduced case the map $\mathcal{Z}$ of [6] realizes the Lascoux–Schützenberger transition on RC graphs. Its K-theoretic counterpart is most naturally described through bumpless pipe dreams. Recall the following ingredients.

(1) **Marked bumpless pipe dreams.** Weigandt [8] introduced *marked bumpless pipe dreams* (marked BPDs) and proved that the double Grothendieck polynomial $\mathfrak{G}_w^{(\beta)}$ is the generating function of marked BPDs of w, extending the Lam–Lee–Shimozono formula from K-theory. Let $\mathrm{MBPD}(w)$ denote the set of marked BPDs of w.
(2) **The HSY bijection.** Huang, Shimozono, and Yu [4] construct a weight-preserving bijection between marked BPDs and reverse compatible pairs, the latter being in bijection with not-necessarily-reduced pipe dreams; their bijection generalizes, in the unmarked reduced case, the canonical bijection of Gao and Huang [3]. Reading a bounded WC graph as the compatible sequence of a (not necessarily reduced) pipe dream, this yields a weight-preserving bijection

$$\Psi : \bigcup_n \{W \in \mathcal{BWC}_n : \mathtt{perm}(W) = w\} \longrightarrow \mathrm{MBPD}(w),$$

whose restriction to RC graphs is the Gao–Huang bijection ϕ used in the reduced theory of [6].
(3) **Weigandt transition.** Weigandt's droop/transition operation $\mathbf{t}$ on bumpless pipe dreams removes the bottom row and rearranges the resulting diagram into a valid BPD of one fewer row, realizing the (co)transition recursion for Grothendieck polynomials; it acts on marked BPDs with an empty bottom row [8, 4].

Definition 4.2. Let W be a bounded WC graph with $\mathtt{ht}(W) = n$ whose row n is empty. Define the *K-theoretic transition map*

$$\mathscr{Z}(W) = \Psi^{-1}\big(\mathbf{t}(\Psi(W))\big),$$

a bounded WC graph of height $n-1$. We extend $\mathscr{Z}$ to an endomorphism of $\mathcal{BWC}$ by setting $\mathscr{Z}(W) = 0$ when row n is nonempty.

The following properties are the K-theoretic analogues of the reduced-case transition lemma, the trim-commutation lemma, and the zeroing bijection of [6]. We record them as the working hypotheses of the construction; each specializes to its reduced counterpart under ρ, where it is a theorem of [6].

Proposition 4.3. *On RC graphs the map $\mathscr{Z}$ agrees with $\mathcal{Z}$: for every bounded WC graph W with empty last row,*

$$\rho(\mathscr{Z}(W)) = \mathcal{Z}(\rho(W)),$$

and in particular $\mathscr{Z}|_{\mathcal{BRC}} = \mathcal{Z}$.

Proof sketch. On RC graphs Ψ is the Gao–Huang bijection ϕ and $\mathbf{t}$ restricts to Weigandt's transition on reduced BPDs, so $\mathscr{Z} = \phi^{-1} \circ \mathbf{t} \circ \phi$ there. The reduced-case identity $\mathcal{Z} \circ \phi^{-1} = \phi^{-1} \circ \mathbf{t}$ (the conjecture of [6] relating $\mathcal{Z}$ and $\mathbf{t}$ through ϕ) gives the claim; ρ discards the non-reduced terms, none of which are produced from a reduced input. $\square$

Proposition 4.4. *Let W be a bounded WC graph with $\mathtt{ht}(W) = n \geq 1$ and empty row n. Then $\mathtt{ht}(\mathscr{Z}(W)) = n-1$, the weight is preserved,*

$$\mathtt{wt}(\mathscr{Z}(W)) = \mathtt{wt}(W),$$

and the permutation undergoes the K-theoretic (Grothendieck) transition: $\mathtt{perm}(\mathscr{Z}(W))$ is obtained from $\mathtt{perm}(W)$ by a single step of the Grothendieck cotransition recursion in row n.

Proposition 4.5. *Let (W, n) be a bounded WC graph with $n \geq 2$ whose row n is empty. Then*

$$\mathscr{Z}(\mathtt{trim}(W)) = \mathtt{trim}(\mathscr{Z}(W)).$$

Proof sketch. This is the K-analogue of the trim-commutation lemma of [6]. Trimming preserves a suffix of the word of W, and both Ψ and $\mathbf{t}$ act on the bottom rows of the marked BPD without reference to the top row: the transition $\mathbf{t}$ proceeds downward from the removed bottom row, so discarding the top row commutes with it. Under ρ the identity specializes to the reduced statement of [6]. $\square$

Proposition 4.6. *Let w be a permutation with last descent at most n, and let $\mathcal{WC}(w,n)^0$ be the set of bounded WC graphs W with $\mathtt{ht}(W) = n$, $\mathtt{perm}(W) = w$, and empty row n. Then*

$$\mathscr{Z} :\ \mathcal{WC}(w,n)^0 \longrightarrow \bigcup_{w \overset{n}{\searrow} w'} \mathcal{WC}(w', n-1)$$

is a weight-preserving bijection, where w' ranges over the outputs of the Grothendieck cotransition in row n. Consequently

$$\mathscr{Z}\Big(\sum_{\substack{\mathtt{perm}(W)=w \\ \mathtt{ht}(W)=n}} W \Big) = \sum_{w \overset{n}{\searrow} w'} \ \sum_{\substack{\mathtt{perm}(W')=w' \\ \mathtt{ht}(W')=n-1}} W',$$

the WC-graph form of the Grothendieck transition formula.

5. WC Graph ring

We now lift the K-theoretic transition map $\mathscr{Z}$ to an associative product on $\mathcal{BWC}$, in exact parallel with the product $\uplus$ on $\mathcal{BRC}$ built from $\mathcal{Z}$ in [6]. Only the underlying zeroing map has changed: $\mathtt{trim}^p$ is unchanged, and $\mathtt{clip}^p$ is now built from $\mathscr{Z}$.

Definition 5.1. Let W be a bounded WC graph with $\mathtt{ht}(W) = n$ and let $0 \leq p \leq n$. Define $\mathtt{trim}^p(W) \in \mathcal{BWC}_{n-p}$ to be the result of discarding the top p rows of W and reindexing (equivalently, the p-fold iterate of $\mathtt{trim}$), and define $\mathtt{clip}^p(W) \in \mathcal{BWC}_p$ to be the $(n-p)$-fold iterate of $\mathscr{Z}$ applied to the bottom row,

$$\mathtt{clip}^p(W) = \mathscr{Z}^{\,n-p}(W),$$

which is defined whenever each intermediate bottom row is empty at the moment it is zeroed (and is 0 otherwise). Thus $\mathtt{clip}^p(W)$ and $\mathtt{trim}^p(W)$ decompose W across the horizontal cut between rows p and $p+1$.

Associativity of the product below rests on two structural identities relating $\mathtt{clip}^p$ and $\mathtt{trim}^p$: functoriality of $\mathtt{clip}$ under iteration, and commutativity of $\mathtt{clip}$ with $\mathtt{trim}$. The first is formal; the second is the content of Proposition 4.5.

Lemma 5.2. *Let W be a bounded WC graph with $\mathtt{ht}(W) = n$ and $0 \leq p \leq p+q \leq n$. Then*

$$\mathtt{clip}^p(\mathtt{clip}^{p+q}(W)) = \mathtt{clip}^p(W).$$

Proof. Both sides are computed by iterating $\mathscr{Z}$ on the bottom row. With $r = n-(p+q)$, the graph $\mathtt{clip}^{p+q}(W)$ is the r-fold iterate $\mathscr{Z}^r(W)$ and $\mathtt{clip}^p(W) = \mathscr{Z}^{r+q}(W)$; the first r iterations in the computation of $\mathtt{clip}^p(W)$ produce $\mathtt{clip}^{p+q}(W)$, and the remaining q produce $\mathtt{clip}^p(\mathtt{clip}^{p+q}(W))$. Hence the two agree. $\square$

Lemma 5.3 (Commutativity of $\mathtt{clip}$ and $\mathtt{trim}$)**.** *Let W be a bounded WC graph with $\mathtt{ht}(W) = p+q+r$. Then*

$$\mathtt{trim}^p(\mathtt{clip}^{p+q}(W)) = \mathtt{clip}^q(\mathtt{trim}^p(W)).$$

Proof. Both sides have height q. By Proposition 4.5, $\mathscr{Z}(\mathtt{trim}(W')) = \mathtt{trim}(\mathscr{Z}(W'))$ whenever the last row of W' is empty. Since $\mathtt{clip}^{p+q}$ on a graph of height $p+q+r$ and $\mathtt{clip}^{q}$ on a graph of height $q+r$ are both the r-fold iterate of $\mathscr{Z}$ on the bottom row (cf. Lemma 5.2), iterating Proposition 4.5 r times gives

$$\mathtt{trim}^p(\mathtt{clip}^{p+q}(W)) = \mathtt{trim}^p(\mathscr{Z}^r(W)) = \mathscr{Z}^r(\mathtt{trim}^p(W)) = \mathtt{clip}^q(\mathtt{trim}^p(W)),$$

as required. □

Definition 5.4. For bounded WC graphs W_1 with $\mathtt{ht}(W_1) = m$ and W_2 with $\mathtt{ht}(W_2) = n$, define

$$W_1 \uplus W_2 = \sum_{\substack{\mathtt{clip}^m(W')=W_1 \\ \mathtt{trim}^m(W')=W_2}} W',$$

where W' ranges over bounded WC graphs with $\mathtt{ht}(W') = m+n$, and extend bilinearly to $\mathcal{BWC}$.

Theorem 5.5. *The product $\uplus$ is associative, and $\mathcal{BWC}$ is a ring under $\uplus$ with identity the empty WC graph of height* 0.

Proof. Let W_1, W_2, W_3 be bounded WC graphs of heights p, q, r, and let W be a bounded WC graph with $\mathtt{ht}(W) = p+q+r$. Unwinding Definition 5.4 in each bracketing, W contributes to the doubly-bracketed products under the conditions

	Left $(W_1 \uplus W_2) \uplus W_3$	Right $W_1 \uplus (W_2 \uplus W_3)$
(i)	$\mathtt{clip}^p(\mathtt{clip}^{p+q}(W)) = W_1$	$\mathtt{clip}^p(W) = W_1$
(ii)	$\mathtt{trim}^p(\mathtt{clip}^{p+q}(W)) = W_2$	$\mathtt{clip}^q(\mathtt{trim}^p(W)) = W_2$
(iii)	$\mathtt{trim}^{p+q}(W) = W_3$	$\mathtt{trim}^q(\mathtt{trim}^p(W)) = W_3$

Row (i) is equivalent on both sides by Lemma 5.2, row (ii) by Lemma 5.3, and row (iii) because $\mathtt{trim}^{p+q} = \mathtt{trim}^q \circ \mathtt{trim}^p$ by definition of $\mathtt{trim}$. Each condition is an equality carrying no combinatorial multiplicity, so W occurs with multiplicity 1 in either bracketing, and the two bracketed products agree. Distributivity is imposed by bilinear extension, and the empty height-0 WC graph is a two-sided identity. □

Because $\mathscr{Z}$ restricts to $\mathcal{Z}$ on RC graphs (Proposition 4.3) and $\mathtt{trim}$ is unchanged, the retraction ρ intertwines the two clip/trim decompositions; hence it is compatible with the products.

Proposition 5.6. *The linear map*

$$\rho : \mathcal{BWC} \to \mathcal{BRC}$$

such that

$$\rho(W) = \begin{cases} W & \text{if } W \text{ is an RC graph} \\ 0 & \text{otherwise} \end{cases}$$

is a surjective ring homomorphism.

Definition 5.7. This homomorphism has a section. Specifically, for each $n > 0$ define a ring Plac_n^K to be the algebra spanned by the n-Grassmannian WC graphs with multiplication given by the squash product. Then we have a surjective ring homomorphism

$$\mathrm{Plac}_n^K \to \mathrm{Plac}_n$$

obtained on elementary symmetric WC graphs by sending $E^{\beta}_{p,k}$ with $|\beta| > p$ to $E^{\beta}_{|\beta|,k}$. Define

$$\mathcal{M}_n^K = \bigotimes_{i=1}^{n} \mathrm{Plac}_i^K$$

There is an evident homomorphism $\mathcal{M}_n^K \to \mathcal{M}_n$ obtained by applying the above homomorphism to each factor. We have a commutative diagram

$$\begin{tikzcd} \mathcal{M}_n^K \arrow[r] \arrow[d] & \mathcal{M}_n \arrow[d] \\ \mathcal{BWC}_n \arrow[r, dashed] & \mathcal{BRC}_n \end{tikzcd}$$

Here the squash product is precisely the restriction of $\uplus$ (Definition 5.4) to n-Grassmannian WC graphs.

Remark. The compatibility of $\uplus$ with the Lascoux decomposition of Theorem 3.8 is deferred. Corollary 3.9 shows that any map preserving the K-Knuth class $\mathsf{P}(W)$ preserves the associated Lascoux polynomial; the outstanding question is how $\mathscr{X}$, and hence $\texttt{clip}^p$ and the squash product, interact with the classes $\mathsf{P}(W)$. We return to this once the properties of $\mathscr{X}$ in Propositions 4.4–4.6 are established.

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